Ex. NO : 10 Date : 27/08/2026

Internetworking with Routers in Cisco Packet tracer simulator.

Aim :

To design & configure an internetwork using routers in cisco packet tracer, connect different networks, assign IP addresses and observe packet transmission.

Procedure :

* Open cisco packet Tracer and create the required network topology.
* create different networks with 3-4 hosts in each network.
* connect the different networks using a router.
* Connect the hosts in each network using the required network devices.
* Assign suitable static IP addresses and subnet masks to all the machines.
* configure the appropriate default gateway for each host.
* Assign suitable IP addresses to the router interfaces connected to the different networks.

*. Activate and verify the required router interfaces.

*. check the IP configuration and connectivity of all the connected devices.

*. perform packet transmission between hosts belonging to different networks.

*. Use Simulation Mode to observe and trace the path followed by the packet through the router.

*. Observe how the router receives and forwards the packet from one network to another.

*. Design & configure an internetwork using a wireless router.

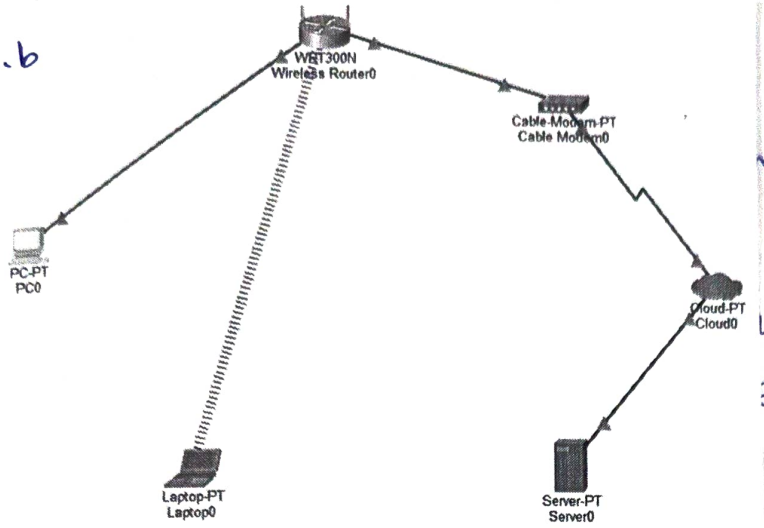
*. Configure the DHCP server in the wireless router to provide IP addresses automatically to connected devices.

*. connect the wireless router network to the internet cloud & verify the connectivity.

*. Design another internetwork using a switch, router & Ethernet cables.

*. connect the required hosts to the switch and connect the switch to the router

10.b

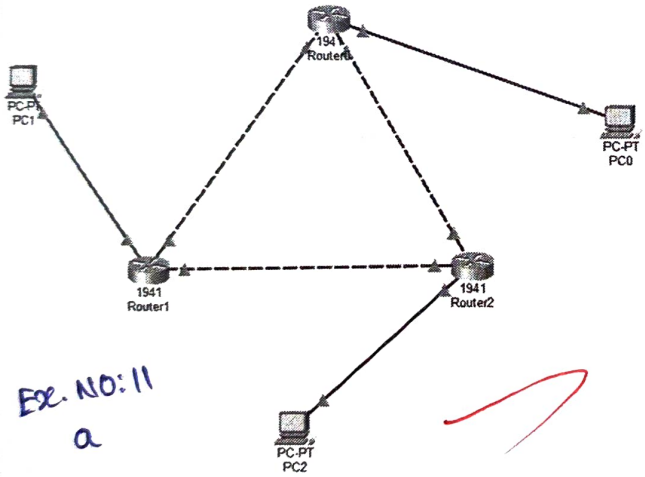


red IP addresses

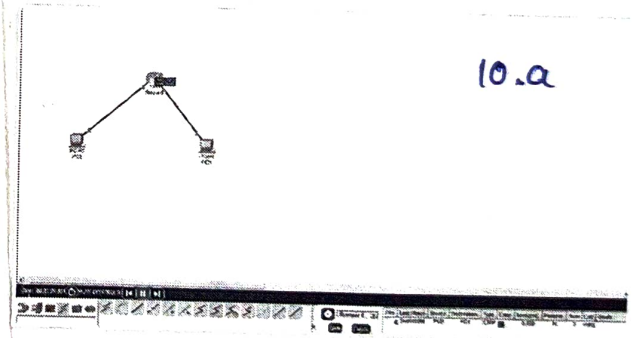
smission & use

the packet

*. Verify that communication is successfully established between the different networks.



Ex. NO: 11
a



10.a

Result:

Thus, an internetwork was successfully designed and configured using routers in Cisco-Packet Tracer. Different networks were connected using routers, static IP addres were assigned, Packet routing was observed, and internetworking using wireless router, DHCP, internet cloud switch & Ethernet cables was successfully verified.